Advanced Computer Networks
Assignment 1-2

Vipul Nayyar (2024CSZ8495)

The version of NS3[1] used to compile and run these tests is '3.45'. The scripts for Part D(*bgp_partD.cc*) and Part E(*bgp_partE.cc*) are placed in the *<NS3>/scratch* directory and then compiled and run as follows from the base NS3 dir:

```
$ ./ns3 build

$NS_LOG="*=prefix_time|prefix_node|prefix_level:Ipv4L3Protocol=level_function|prefix_time|prefix_node|prefix_level" ./build/scratch/ns3.45-bgp_partD-default 2>&1 | grep -Ev "Ipv4L3Protocol:Send|Ipv4L3Protocol:BuildHeader|Ipv4L3Protocol:LocalDeliver|Ipv4L3Protocol:Receive"

$NS_LOG="*=prefix_time|prefix_node|prefix_level:Ipv4L3Protocol=level_function|prefix_time|prefix_node|prefix_level" ./build/scratch/ns3.45-bgp_partE-default 2>&1 | grep -Ev "Ipv4L3Protocol:Send|Ipv4L3Protocol:BuildHeader|Ipv4L3Protocol:LocalDeliver|Ipv4L3Protocol:Receive"
```

Part D: Failure and recovery study

This test consists of 3 nodes/routers, along with their associated interface IP addresses, as shown in Figure 1.

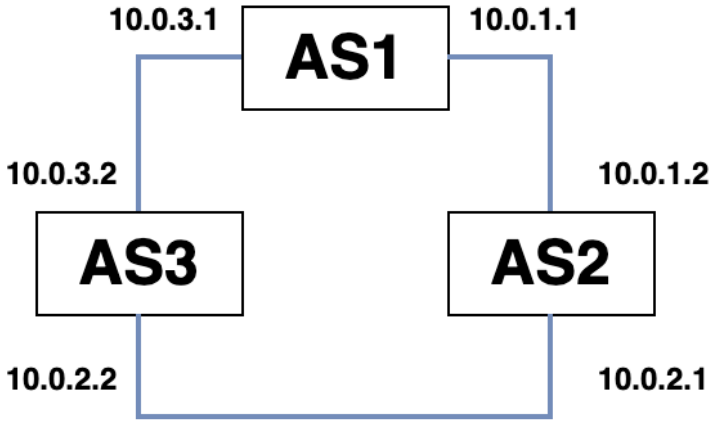


Figure 1: Topology of AS1-3 with their interface IP addresses

In this test, the client on AS1 sends a packet from **10.0.1.1** to the server on AS3 at **10.0.2.2**, which responds with a packet back to the client. A link failure is introduced on the AS1-AS3 link by bringing down both the 10.0.3.1 interface on AS1 and the 10.0.3.2 interface on AS3 together at t = 5s. Both these interfaces are brought back up at t = 7s. The following instructions in the NS3[1] script are used.

```
void StopLinkInterface(Ptr<Node> router1, int32_t interface_index1, Ptr<Node> router2, int32_t
interface_index2){
  router1->GetObject<Ipv4>()->SetDown(interface_index1);
  router2->GetObject<Ipv4>()->SetDown(interface_index2);
}

void StartLinkInterface(Ptr<Node> router1, int32_t interface_index1, Ptr<Node> router2, int32_t
interface_index2){
 router1->GetObject<Ipv4>()->SetUp(interface_index1);
 router2->GetObject<Ipv4>()->SetUp(interface_index2);
}
```

This setup uses the Ad Hoc On-Demand Distance Vector (AODV)[2] protocol for dynamic routing.

```
121  +4.000000000s 0 [INFO ] At time +4s client sent 1024 bytes to 10.0.2.2 port 9
122  +4.002843200s 2 [INFO ] At time +4.00284s server received 1024 bytes from 10.0.3.1 port 49153
123  +4.002843200s 2 [INFO ] At time +4.00284s server sent 1024 bytes to 10.0.3.1 port 49153
124  +4.005686400s 0 [INFO ] At time +4.00569s client received 1024 bytes from 10.0.3.2 port 9
125  +5.000000000s -1 Ipv4L3Protocol:SetDown(0x125e05060, 2)
126  +5.000000000s -1 Ipv4L3Protocol:SetDown(0x125e05960, 2)
127  +5.000000000s 0 [INFO ] At time +5s client sent 1024 bytes to 10.0.2.2 port 9
128  +5.013083200s 0 Ipv4L3Protocol:IpForward(0x125e05060, 0x60000073f540, 0x600002214150, tos 0x0 DSCP
     Default ECN Not-ECT ttl 65 id 0 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.1.1 > 10.0.
     2.2)
129  +5.015952000s 1 Ipv4L3Protocol:IpForward(0x125e05290, 0x60000074c540, 0x600002214070, tos 0x0 DSCP
     Default ECN Not-ECT ttl 64 id 0 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.1.1 > 10.0.
     2.2)
130  +5.018795200s 2 [INFO ] At time +5.0188s server received 1024 bytes from 10.0.1.1 port 49153
131  +5.018795200s 2 [INFO ] At time +5.0188s server sent 1024 bytes to 10.0.1.1 port 49153
132  +5.021638400s 1 Ipv4L3Protocol:IpForward(0x125e05290, 0x6000007481e0, 0x6000022144d0, tos 0x0 DSCP
     Default ECN Not-ECT ttl 64 id 1 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.2.2 > 10.0.
     1.1)
133  +5.024481600s 0 [INFO ] At time +5.02448s client received 1024 bytes from 10.0.2.2 port 9
134  +6.000000000s 0 [INFO ] At time +6s client sent 1024 bytes to 10.0.2.2 port 9
135  +6.002843200s 1 Ipv4L3Protocol:IpForward(0x125e05290, 0x60000074c540, 0x600002214540, tos 0x0 DSCP
     Default ECN Not-ECT ttl 64 id 1 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.1.1 > 10.0.
     2.2)
136  +6.005686400s 2 [INFO ] At time +6.00569s server received 1024 bytes from 10.0.1.1 port 49153
137  +6.005686400s 2 [INFO ] At time +6.00569s server sent 1024 bytes to 10.0.1.1 port 49153
138  +6.008529600s 1 Ipv4L3Protocol:IpForward(0x125e05290, 0x600000748120, 0x600002214150, tos 0x0 DSCP
     Default ECN Not-ECT ttl 64 id 2 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.2.2 > 10.0.
     1.1)
139  +6.011372800s 0 [INFO ] At time +6.01137s client received 1024 bytes from 10.0.2.2 port 9
140  +7.000000000s -1 Ipv4L3Protocol:SetUp(0x125e05060, 2)
141  +7.000000000s -1 Ipv4L3Protocol:SetUp(0x125e05960, 2)
142  +7.000000000s 0 [INFO ] At time +7s client sent 1024 bytes to 10.0.2.2 port 9
143  +7.002843200s 1 Ipv4L3Protocol:IpForward(0x125e05290, 0x60000074c540, 0x6000022140e0, tos 0x0 DSCP
     Default ECN Not-ECT ttl 64 id 2 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.1.1 > 10.0.
     2.2)
144  +7.005686400s 2 [INFO ] At time +7.00569s server received 1024 bytes from 10.0.1.1 port 49153
145  +7.005686400s 2 [INFO ] At time +7.00569s server sent 1024 bytes to 10.0.1.1 port 49153
146  +7.008529600s 1 Ipv4L3Protocol:IpForward(0x125e05290, 0x600000748120, 0x6000022143f0, tos 0x0 DSCP
     Default ECN Not-ECT ttl 64 id 3 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.2.2 > 10.0.
     1.1)
147  +7.011372800s 0 [INFO ] At time +7.01137s client received 1024 bytes from 10.0.2.2 port 9
```

Figure 2: Console excerpt showing route change on AS1-AS3 link failure. First entry in every log line is simulated time in seconds, and the second entry is the node ID(node0 -> AS1, node1 -> AS2, node2 -> AS3)

As per the console excerpt in Figure 2 from the NS3 run, it can be seen that before the failure at t=4s(Line 121 - 124), the client packet travels from AS1 to AS3(node 0 -> node 2), and returns in the reverse path. At t=5s(Line 125 - 133), when a link failure between AS1 and AS3 is simulated, the packet is sent using an alternate path from AS1->AS2->AS3(node 0 -> node1 -> node2) because of the dynamic routing change implemented by the AODV protocol.

At t=7s, when the failed link is restored, the packets keep on travelling over the new path(AS1-AS2-AS3) until the end of the simulation. Figures 3, 4 and 5 show the routing tables for all nodes before the link failure(t=4s), during the failure(t=6s), and after the failure has recovered(t=8s). It can be observed that the AS1-AS3 route was used before failure to get to **10.0.2.2.** During the failure, the routing protocol used an alternate route AS1 -> AS2 -> AS3. Finally, after the failure this newly computed route remains in the routing table and is followed as can be verified from the log(*partD/bgp_partD.log*).

```
Node: 0, Time: +4s, Local time: +4s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 0; Time: +4s, Local time: +4s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.2        10.0.1.2        10.0.1.1        UP              +1s             1
10.0.1.255      10.0.1.255      10.0.1.1        UP              +9.2e+09s       1
10.0.2.2        10.0.3.2        10.0.3.1        UP              +9.2s           1
10.0.3.2        10.0.3.2        10.0.3.1        UP              +2s             1
10.0.3.255      10.0.3.255      10.0.3.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 1, Time: +4s, Local time: +4s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 1; Time: +4s, Local time: +4s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.1.1        10.0.1.2        UP              +2s             1
10.0.1.255      10.0.1.255      10.0.1.2        UP              +9.2e+09s       1
10.0.2.2        10.0.2.2        10.0.2.1        UP              +1s             1
10.0.2.255      10.0.2.255      10.0.2.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 2, Time: +4s, Local time: +4s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 2; Time: +4s, Local time: +4s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.2.1        10.0.2.2        UP              +1s             2
10.0.2.1        10.0.2.1        10.0.2.2        UP              +1s             1
10.0.2.255      10.0.2.255      10.0.2.2        UP              +9.2e+09s       1
10.0.3.1        10.0.3.1        10.0.3.2        UP              +2s             1
10.0.3.255      10.0.3.255      10.0.3.2        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1
```

Figure 3: Routing table for all nodes before link failure at t=4s

```
Node: 0, Time: +6s, Local time: +6s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 0; Time: +6s, Local time: +6s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.2        10.0.1.2        10.0.1.1        UP              +2s             1
10.0.1.255      10.0.1.255      10.0.1.1        UP              +9.2e+09s       1
10.0.2.2        10.0.1.2        10.0.1.1        UP              +0.012s         2
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 1, Time: +6s, Local time: +6s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 1; Time: +6s, Local time: +6s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.1.1        10.0.1.2        UP              +2s             1
10.0.1.255      10.0.1.255      10.0.1.2        UP              +9.2e+09s       1
10.0.2.2        10.0.2.2        10.0.2.1        UP              +2s             1
10.0.2.255      10.0.2.255      10.0.2.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 2, Time: +6s, Local time: +6s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 2; Time: +6s, Local time: +6s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.2.1        10.0.2.2        UP              +2s             2
10.0.2.1        10.0.2.1        10.0.2.2        UP              +2s             1
10.0.2.255      10.0.2.255      10.0.2.2        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1
```

Figure 4: Routing table for all nodes during link failure at t=6s, when the routing protocol has updated the routes

```
Node: 0, Time: +8s, Local time: +8s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 0; Time: +8s, Local time: +8s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.2        10.0.1.2        10.0.1.1        UP              +2s             1
10.0.1.255      10.0.1.255      10.0.1.1        UP              +9.2e+09s       1
10.0.2.2        10.0.1.2        10.0.1.1        UP              +2s             2
10.0.3.2        10.0.3.2        10.0.3.1        UP              +2s             1
10.0.3.255      10.0.3.255      10.0.3.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 1, Time: +8s, Local time: +8s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 1; Time: +8s, Local time: +8s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.1.1        10.0.1.2        UP              +2s             1
10.0.1.255      10.0.1.255      10.0.1.2        UP              +9.2e+09s       1
10.0.2.2        10.0.2.2        10.0.2.1        UP              +2s             1
10.0.2.255      10.0.2.255      10.0.2.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 2, Time: +8s, Local time: +8s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 2; Time: +8s, Local time: +8s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.2.1        10.0.2.2        UP              +2s             2
10.0.2.1        10.0.2.1        10.0.2.2        UP              +2s             1
10.0.2.255      10.0.2.255      10.0.2.2        UP              +9.2e+09s       1
10.0.3.1        10.0.3.1        10.0.3.2        UP              +2s             1
10.0.3.255      10.0.3.255      10.0.3.2        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1
```

Figure 5: Routing table for all nodes after recovery from the link failure at t=8s

For flow analysis, three different flow dumps are taken during the run. First, just when failure occurs at t=5s, second at t=7s when failure ends, and the third at the end of the simulation.

Phase	Packets delivered	Packets lost	Average Delay(ms)
Before failure(t = 0 to t = 5s)	9	0	2.788
During failure(t = 5s to t = 7s)	7	0	3.0041
After failure(t = 7s to end of simulation)	6	0	1.4215

Table 1: Flow analysis for Part D

Part E: Dual-path rerouting study

This test consists of 4 routers R1-R4(node0-node3 in NS3), using the AODV dynamic routing protocol.

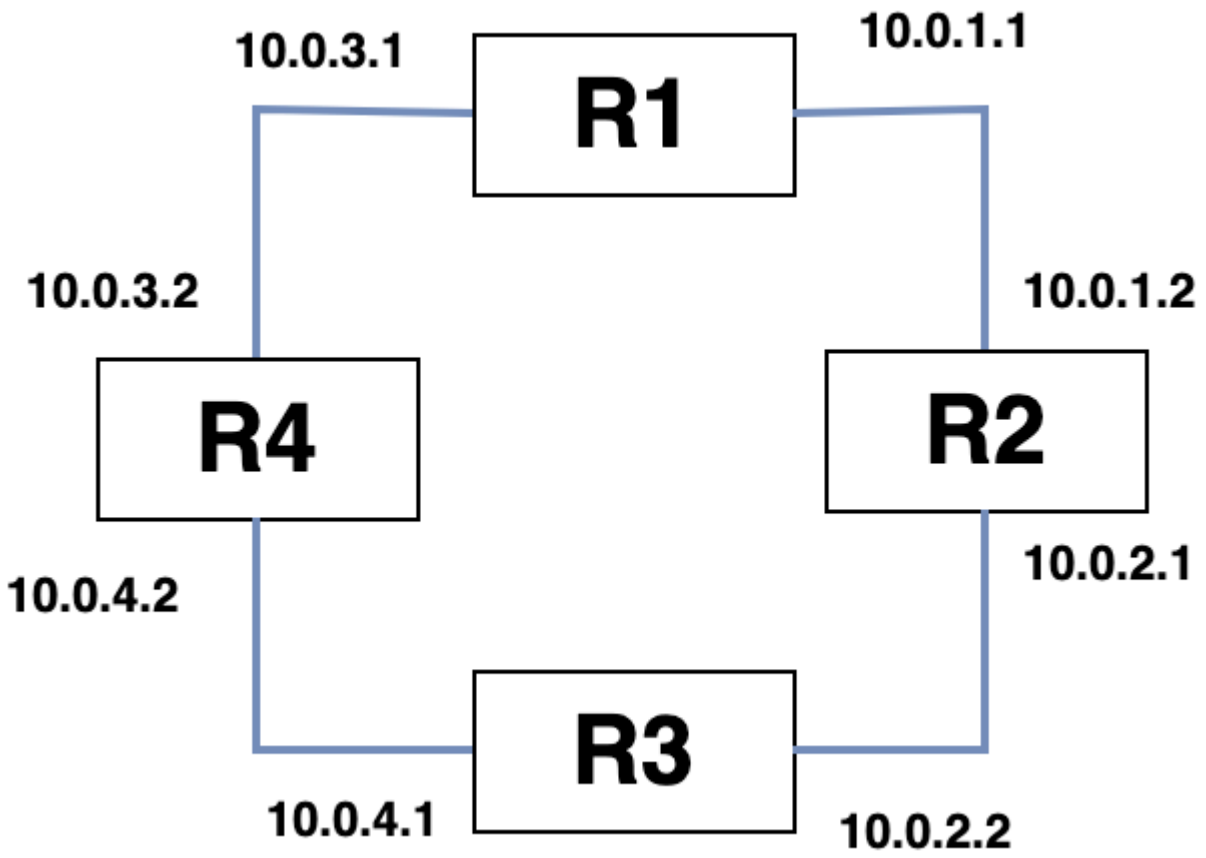


Figure 6: Topology of R1-R4 with their associated interface IP addresses.

In this test, the client on R1(**10.0.1.1**) sends a packet to the server on R3(**10.0.2.2**), which sends a response packet back to the client. A link failure is introduced on R2-R3 by bringing down the interface(10.0.2.1) on R2 with the following instructions.

```
void StopLinkInterface(Ptr<Node> router1, int32_t interface_index1){
  router1->GetObject<Ipv4>()->SetDown(interface_index1);
}

void StartLinkInterface(Ptr<Node> router1, int32_t interface_index1){
 router1->GetObject<Ipv4>()->SetUp(interface_index1);
}
```

```
159 +4.000000000s 0 [INFO ] At time +4s client sent 1024 bytes to 10.0.2.2 port 9
160 +4.002843200s 1 Ipv4L3Protocol:IpForward(0x12a61d180, 0x600003118480, 0x60000146c000, tos 0x0 DSCP Default ECN Not-ECT
    ttl 64 id 2 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.1.1 > 10.0.2.2)
161 +4.005686400s 2 [INFO ] At time +4.00569s server received 1024 bytes from 10.0.1.1 port 49153
162 +4.005686400s 2 [INFO ] At time +4.00569s server sent 1024 bytes to 10.0.1.1 port 49153
163 +4.008529600s 1 Ipv4L3Protocol:IpForward(0x12a61d180, 0x600003114cf0, 0x60000146c070, tos 0x0 DSCP Default ECN Not-ECT
    ttl 64 id 2 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.2.2 > 10.0.1.1)
164 +4.011372800s 0 [INFO ] At time +4.01137s client received 1024 bytes from 10.0.2.2 port 9
165 +5.000000000s -1 Ipv4L3Protocol:SetDown(0x12a61d180, 2)
166 +5.000000000s 0 [INFO ] At time +5s client sent 1024 bytes to 10.0.2.2 port 9
167 +5.002843200s 1 [WARN ] No route found for forwarding packet.  Drop.
168 +6.000000000s 0 [INFO ] At time +6s client sent 1024 bytes to 10.0.2.2 port 9
169 +6.023206400s 0 Ipv4L3Protocol:IpForward(0x11a604080, 0x60000311dfe0, 0x60000147c310, tos 0x0 DSCP Default ECN Not-ECT
    ttl 65 id 4 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.3.1 > 10.0.2.2)
170 +6.026049600s 3 Ipv4L3Protocol:IpForward(0x11a804640, 0x60000311d620, 0x60000147f170, tos 0x0 DSCP Default ECN Not-ECT
    ttl 64 id 4 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.3.1 > 10.0.2.2)
171 +6.028892800s 2 [INFO ] At time +6.02889s server received 1024 bytes from 10.0.3.1 port 49153
172 +6.028892800s 2 [INFO ] At time +6.02889s server sent 1024 bytes to 10.0.3.1 port 49153
173 +6.031736000s 3 Ipv4L3Protocol:IpForward(0x11a804640, 0x600003114cc0, 0x60000147f100, tos 0x0 DSCP Default ECN Not-ECT
    ttl 64 id 0 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.4.1 > 10.0.3.1)
174 +6.034579200s 0 [INFO ] At time +6.03458s client received 1024 bytes from 10.0.4.1 port 9
175 +7.000000000s -1 Ipv4L3Protocol:SetUp(0x12a61d180, 2)
176 +7.000000000s 0 [INFO ] At time +7s client sent 1024 bytes to 10.0.2.2 port 9
177 +7.002843200s 3 Ipv4L3Protocol:IpForward(0x11a804640, 0x60000311d620, 0x60000147efb0, tos 0x0 DSCP Default ECN Not-ECT
    ttl 64 id 0 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.3.1 > 10.0.2.2)
178 +7.005686400s 2 [INFO ] At time +7.00569s server received 1024 bytes from 10.0.3.1 port 49153
179 +7.005686400s 2 [INFO ] At time +7.00569s server sent 1024 bytes to 10.0.3.1 port 49153
180 +7.008529600s 3 Ipv4L3Protocol:IpForward(0x11a804640, 0x600003114cc0, 0x60000147c2a0, tos 0x0 DSCP Default ECN Not-ECT
    ttl 64 id 1 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.4.1 > 10.0.3.1)
181 +7.011372800s 0 [INFO ] At time +7.01137s client received 1024 bytes from 10.0.4.1 port 9
182 +8.000000000s 0 [INFO ] At time +8s client sent 1024 bytes to 10.0.2.2 port 9
183 +8.002843200s 3 Ipv4L3Protocol:IpForward(0x11a804640, 0x60000311d620, 0x60000146c0e0, tos 0x0 DSCP Default ECN Not-ECT
    ttl 64 id 1 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.3.1 > 10.0.2.2)
184 +8.005686400s 2 [INFO ] At time +8.00569s server received 1024 bytes from 10.0.3.1 port 49153
185 +8.005686400s 2 [INFO ] At time +8.00569s server sent 1024 bytes to 10.0.3.1 port 49153
186 +8.008529600s 3 Ipv4L3Protocol:IpForward(0x11a804640, 0x60000317ee20, 0x60000146c000, tos 0x0 DSCP Default ECN Not-ECT
    ttl 64 id 2 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.4.1 > 10.0.3.1)
187 +8.011372800s 0 [INFO ] At time +8.01137s client received 1024 bytes from 10.0.4.1 port 9
```

Figure 7: Console excerpt showing route change on R2-R3 link failure. First entry in every log line is simulated time in seconds, and the second entry is the node ID(node0 -> R1, node1 -> R2, node2 -> R3, node3 -> R4)

According to the console excerpt in Figure 7, at t=4s(Lines 159-164), before the failure the packet travels from R1->R2->R3(node0->node1->node2) and back in reverse. During the link failure at t=5s, the client packet is dropped at R2(node1) in Line 167, as the dynamic routing protocol requires time to update and converge the new routes across all nodes. At t=6s, during the failure(Lines 168-174), route convergence is complete, and the alternate route is used from R1->R4->R3(node0->node3->node2), and the reverse while going back. After the failure is restored at t =7s, it can be seen in Lines 176-187 that the modified routes are again used till the end of the simulation.

Figures 8, 9, 10, and 11 show the routing tables for all nodes at t=4s, t=5s, t=6.5s, and t=8s. It can be seen in FIgure 8 that the route for R3(**10.0.2.2**) is present in R1(node 0) and R2(Node1) routing tables at t=4s. During failure at t=5s in Figure 9, this route for R3 is missing from R2, but present in R1(node0). This leads to the client packet being dropped at R2 as seen on Line 167 in Figure 7. At t=6.5s in Figure 10, the route for R3 is now updated in R1(node 0) and added to R4(node 3), which leads to the client packet now taking this alternate path(R1->R4->R3). Finally, at t=8s in Figure 11, when the failure is over, the updates routes persist and the client packets continue taking this new path.

```
Node: 0, Time: +4s, Local time: +4s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 0; Time: +4s, Local time: +4s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.2        10.0.1.2        10.0.1.1        UP              +2s             1
10.0.1.255      10.0.1.255      10.0.1.1        UP              +9.2e+09s       1
10.0.2.2        10.0.1.2        10.0.1.1        UP              +2s             2
10.0.3.2        10.0.3.2        10.0.3.1        UP              +1s             1
10.0.3.255      10.0.3.255      10.0.3.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 1, Time: +4s, Local time: +4s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 1; Time: +4s, Local time: +4s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.1.1        10.0.1.2        UP              +2s             1
10.0.1.255      10.0.1.255      10.0.1.2        UP              +9.2e+09s       1
10.0.2.2        10.0.2.2        10.0.2.1        UP              +2s             1
10.0.2.255      10.0.2.255      10.0.2.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 2, Time: +4s, Local time: +4s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 2; Time: +4s, Local time: +4s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.2.1        10.0.2.2        UP              +2s             2
10.0.2.1        10.0.2.1        10.0.2.2        UP              +2s             1
10.0.2.255      10.0.2.255      10.0.2.2        UP              +9.2e+09s       1
10.0.4.2        10.0.4.2        10.0.4.1        UP              +1s             1
10.0.4.255      10.0.4.255      10.0.4.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 3, Time: +4s, Local time: +4s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 3; Time: +4s, Local time: +4s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.3.1        10.0.3.1        10.0.3.2        UP              +2s             1
10.0.3.255      10.0.3.255      10.0.3.2        UP              +9.2e+09s       1
10.0.4.1        10.0.4.1        10.0.4.2        UP              +1s             1
10.0.4.255      10.0.4.255      10.0.4.2        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1
```

Figure 8: Routing table for all nodes before link failure at t=4s

```
Node: 0, Time: +5s, Local time: +5s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 0; Time: +5s, Local time: +5s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.2        10.0.1.2        10.0.1.1        UP              +2s             1
10.0.1.255      10.0.1.255      10.0.1.1        UP              +9.2e+09s       1
10.0.2.2        10.0.1.2        10.0.1.1        UP              +2s             2
10.0.3.2        10.0.3.2        10.0.3.1        UP              +1s             1
10.0.3.255      10.0.3.255      10.0.3.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 1, Time: +5s, Local time: +5s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 1; Time: +5s, Local time: +5s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.1.1        10.0.1.2        UP              +2s             1
10.0.1.255      10.0.1.255      10.0.1.2        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 2, Time: +5s, Local time: +5s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 2; Time: +5s, Local time: +5s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.2.1        10.0.2.2        UP              +2s             2
10.0.2.1        10.0.2.1        10.0.2.2        UP              +2s             1
10.0.2.255      10.0.2.255      10.0.2.2        UP              +9.2e+09s       1
10.0.4.2        10.0.4.2        10.0.4.1        UP              +1s             1
10.0.4.255      10.0.4.255      10.0.4.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 3, Time: +5s, Local time: +5s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 3; Time: +5s, Local time: +5s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.3.1        10.0.3.1        10.0.3.2        UP              +1s             1
10.0.3.255      10.0.3.255      10.0.3.2        UP              +9.2e+09s       1
10.0.4.1        10.0.4.1        10.0.4.2        UP              +1s             1
10.0.4.255      10.0.4.255      10.0.4.2        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1
```

Figure 9: Routing table for all nodes during link failure at t=5s, before alternate route update, and convergence is pending

```
Node: 0, Time: +6.5s, Local time: +6.5s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 0; Time: +6.5s, Local time: +6.5s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.2        10.0.1.2        10.0.1.1        UP              +1.5s           1
10.0.1.255      10.0.1.255      10.0.1.1        UP              +9.2e+09s       1
10.0.2.2        10.0.3.2        10.0.3.1        UP              +11s            2
10.0.3.2        10.0.3.2        10.0.3.1        UP              +0.54s          1
10.0.3.255      10.0.3.255      10.0.3.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 1, Time: +6.5s, Local time: +6.5s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 1; Time: +6.5s, Local time: +6.5s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.1.1        10.0.1.2        UP              +2.5s           1
10.0.1.255      10.0.1.255      10.0.1.2        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 2, Time: +6.5s, Local time: +6.5s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 2; Time: +6.5s, Local time: +6.5s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.2.1        10.0.2.2        UP              +0.51s          2
10.0.2.1        10.0.2.1        10.0.2.2        UP              +0.51s          1
10.0.2.255      10.0.2.255      10.0.2.2        UP              +9.2e+09s       1
10.0.3.1        10.0.4.2        10.0.4.1        UP              +5s             2
10.0.4.2        10.0.4.2        10.0.4.1        UP              +2.5s           1
10.0.4.255      10.0.4.255      10.0.4.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 3, Time: +6.5s, Local time: +6.5s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 3; Time: +6.5s, Local time: +6.5s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.2.2        10.0.4.1        10.0.4.2        UP              +11s            1
10.0.3.1        10.0.3.1        10.0.3.2        UP              +2.5s           1
10.0.3.255      10.0.3.255      10.0.3.2        UP              +9.2e+09s       1
10.0.4.1        10.0.4.1        10.0.4.2        UP              +2.5s           1
10.0.4.255      10.0.4.255      10.0.4.2        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1

```

Figure 10: Routing table for all nodes during link failure at t=6.5s, when the alternate route update and convergence are complete

```
Node: 0, Time: +8s, Local time: +8s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 0; Time: +8s, Local time: +8s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.2        10.0.1.2        10.0.1.1        UP              +1s             1
10.0.1.255      10.0.1.255      10.0.1.1        UP              +9.2e+09s       1
10.0.2.2        10.0.3.2        10.0.3.1        UP              +9.2s           2
10.0.3.2        10.0.3.2        10.0.3.1        UP              +2s             1
10.0.3.255      10.0.3.255      10.0.3.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 1, Time: +8s, Local time: +8s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 1; Time: +8s, Local time: +8s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.1.1        10.0.1.2        UP              +2s             1
10.0.1.255      10.0.1.255      10.0.1.2        UP              +9.2e+09s       1
10.0.2.2        10.0.2.2        10.0.2.1        UP              +2s             1
10.0.2.255      10.0.2.255      10.0.2.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 2, Time: +8s, Local time: +8s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 2; Time: +8s, Local time: +8s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.1.1        10.0.2.1        10.0.2.2        DOWN            +14s            2
10.0.2.1        10.0.2.1        10.0.2.2        UP              +14s            1
10.0.2.255      10.0.2.255      10.0.2.2        UP              +9.2e+09s       1
10.0.3.1        10.0.4.2        10.0.4.1        UP              +3.5s           2
10.0.4.2        10.0.4.2        10.0.4.1        UP              +2s             1
10.0.4.255      10.0.4.255      10.0.4.1        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1


Node: 3, Time: +8s, Local time: +8s, Ipv4ListRouting table
  Priority: 0 Protocol: ns3::aodv::RoutingProtocol
Node: 3; Time: +8s, Local time: +8s, AODV Routing table

AODV Routing table
Destination     Gateway         Interface       Flag            Expire          Hops
10.0.2.2        10.0.4.1        10.0.4.2        UP              +9.2s           1
10.0.3.1        10.0.3.1        10.0.3.2        UP              +2s             1
10.0.3.255      10.0.3.255      10.0.3.2        UP              +9.2e+09s       1
10.0.4.1        10.0.4.1        10.0.4.2        UP              +2s             1
10.0.4.255      10.0.4.255      10.0.4.2        UP              +9.2e+09s       1
127.0.0.1       127.0.0.1       127.0.0.1       UP              +9.2e+09s       1
```

Figure 11: Routing table for all nodes during link failure at t=8s, after the failure has recovered

For flow analysis[3], three different flow dumps are taken during the run. First, just when failure occurs at t=5s, second at t=7s when failure ends, and third at the end of the simulation. Table 2 presents the results from Flowmonitor[3].

Phase	Packets delivered	Packets lost	Average Delay(ms)
Before failure(t = 0 to t = 5s)	9	0	4.436
During failure(t = 5s to t = 7s)	4	1	1.475
After failure(t = 7s to end of simulation)	6	0	1.263

Table 2: Flow analysis for Part E

In this test, the following route changes happen according to Table 3.

Before/After	Router	Route to R3
Before (Figure 8)	R1	10.0.2.2 via 10.0.1.2(R2)
	R2	10.0.2.2 via 10.0.2.2 (R3 direct)
	R3	Self(No Route)
	R4	No Route
After (Figure 11)	R1	10.0.2.2 via 10.0.3.2(R4)
	R2	10.0.2.2 via 10.0.2.2 (R3 direct)
	R3	Self(No Route)
	R4	10.0.2.2 via 10.0.4.1 (R3 direct)

Table 3: Routing changes in each node pointing to R3(10.0.2.2) from t=4s to t=8s after failure recovery

References

1) https://www.nsnam.org/
2) https://www.nsnam.org/docs/models/html/aodv.html
3) https://www.nsnam.org/docs/models/html/flow-monitor.html